

# Data Mining

## Clustering: Part 3

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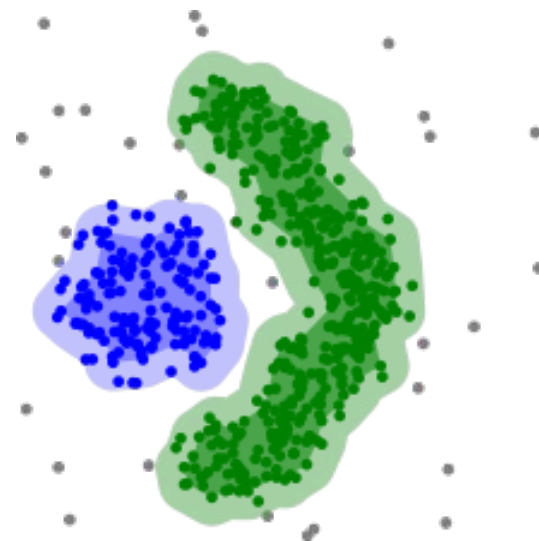
# Outline

- ❑ **Overview of Clustering**
- ❑ **Major Clustering Approaches**
  - ❑ **K-means Clustering**
  - ❑ **Hierarchical Clustering**
  - ❑ **DBSCAN Clustering**
- ❑ **Cluster Evaluation**

# Density-based Clustering

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- Density-based methods use **density** to discover clusters of any shape.
- Density means the **concentration** of data points in a given region.
- Clusters are regions of **high density** separated by regions of **low density**.
- Major features of density-based clustering:
  - Discover clusters of arbitrary shape
  - Handle noise and outliers
  - One scan only
  - No need to define the number of clusters in advance



# DBSCAN Clustering

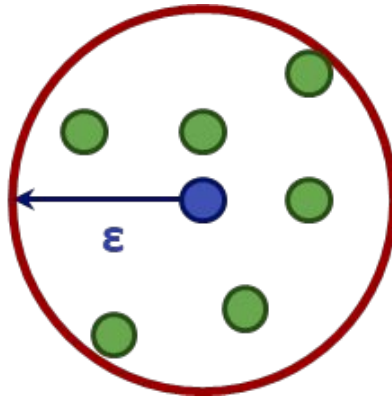
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- DBSCAN: **D**ensity-**B**ased **S**patial **C**lustering of **A**pplications with **N**oise
- It defines a cluster as a **maximal set of density-connected points**.
- **Density** is the number of data points within a certain **radius**.
- The parameters of this algorithm are:
  - $\epsilon$ : the maximum radius of the neighborhood
  - **MinPts**: the minimal number of point in a dense neighborhood.

# $\epsilon$ -neighborhood

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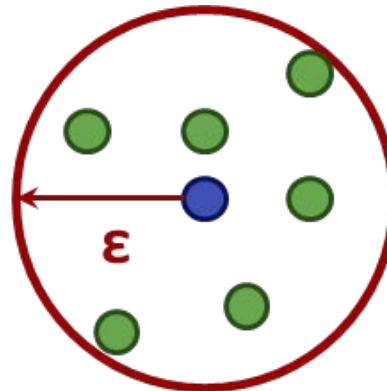
- The neighborhood within a radius  $\epsilon$  of a given object is called the  **$\epsilon$ -neighborhood** of the object.



# Core Objects

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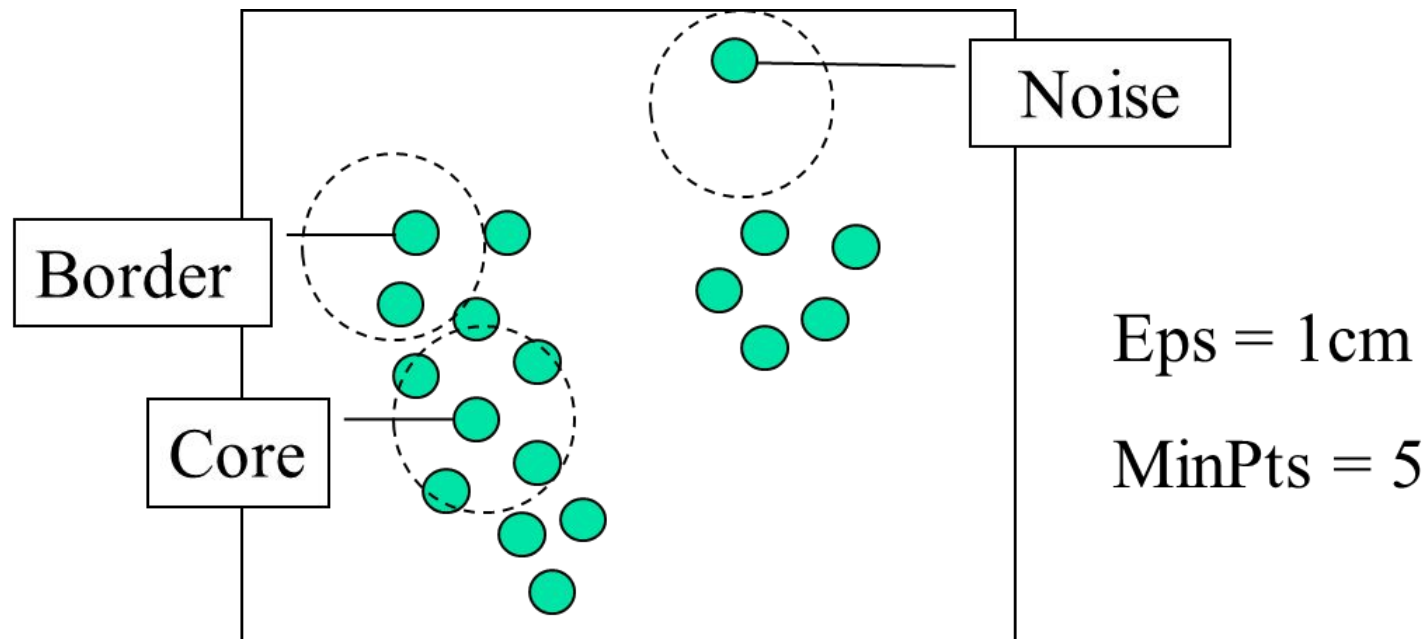
- If the  **$\epsilon$ -neighborhood** of an object contains at least a minimum number of objects **MinPts**, then the object is called a **core object**.



MinPts = 5

# Core, Border, and Noise Points

- A **core point** has at least a specified number of points (**MinPts**) within  $\epsilon$
- A **border point** is not a core point, but is in the neighborhood of a core point
- A **noise point** is any point that is not a core point or a border point

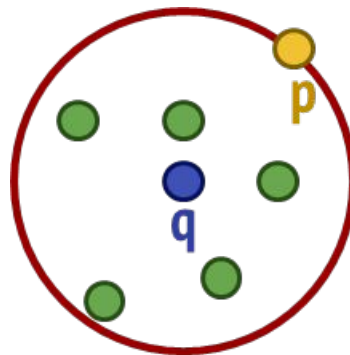


# Density Reachability (1 / 2)

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- Given a set of objects  $\mathbf{D}$ , we say that an object  $p$  is **directly density-reachable** from an object  $q$  if:

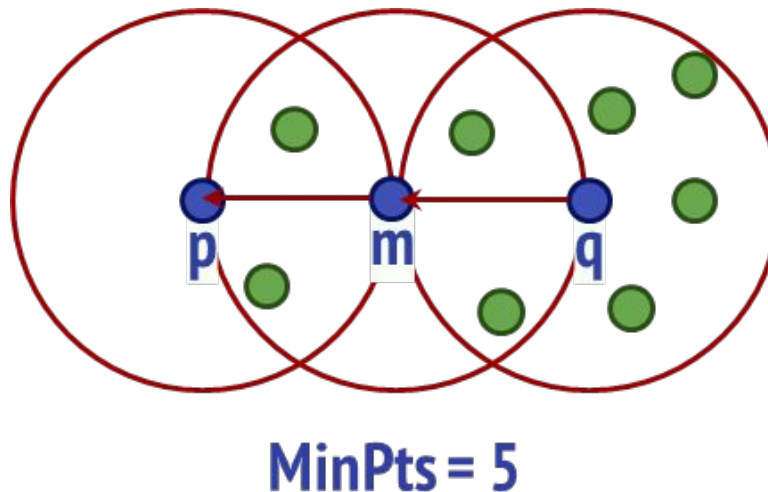
“ $p$  is within the  $\epsilon$ -neighborhood of  $q$ , and  $q$  is a **core object**.”



# Density Reachability (2 / 2)

- Given a set of objects  $\mathbf{D}$ , an object  $\mathbf{p}$  is **density-reachable** from an object  $\mathbf{q}$  with respect to  $\epsilon$  and **MinPts** if:

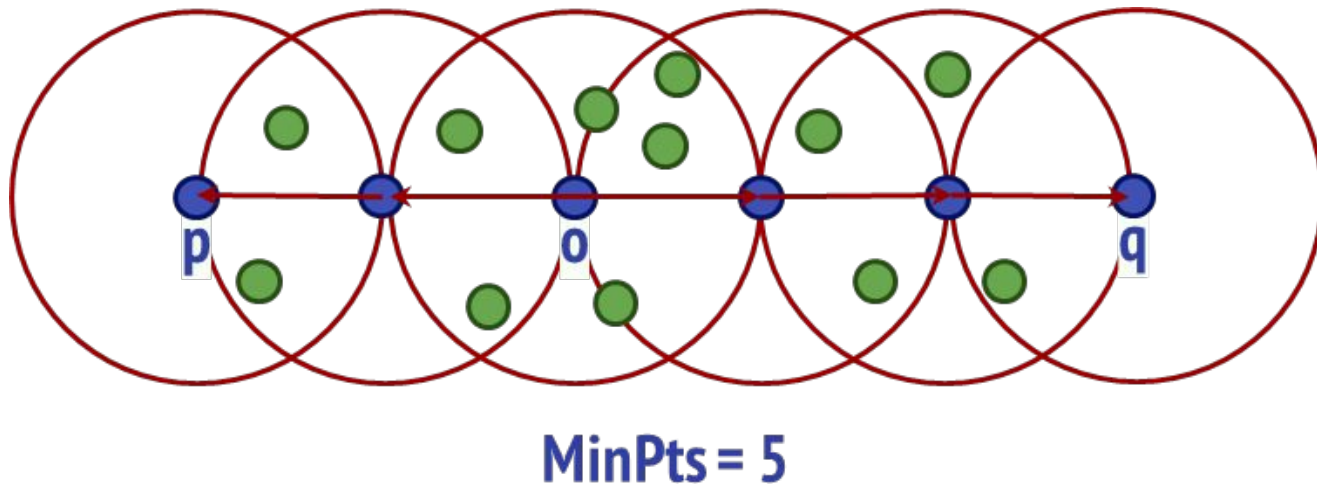
“There is a chain of objects  $\mathbf{p}_1, \dots, \mathbf{p}_n$ , where  $\mathbf{p}_1 = \mathbf{q}$  and  $\mathbf{p}_n = \mathbf{p}$  such that  $\mathbf{p}_{i+1}$  is **directly density-reachable** from  $\mathbf{p}_i$  with respect to  $\epsilon$  and **MinPts**, for  $1 \leq i \leq n$  and  $\mathbf{p}_i \in \mathbf{D}$ .”



# Density Connectivity

- Given a set of objects **D**, an object **p** is **density-connected** to an object **q** with respect to  $\epsilon$  and **MinPts** if:

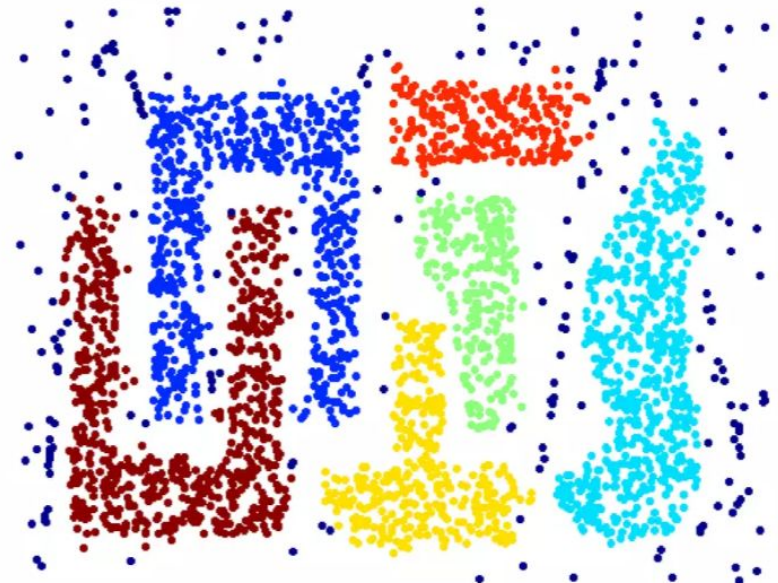
“There is an object **o**  $\in$  **D** such that both **p** and **q** are **density-reachable** from **o** with respect to  $\epsilon$  and **MinPts**”



# Density-based Clusters

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- A **density-based cluster** is a set of **density-connected** points that is maximal with respect to **density-reachability**.
- Every object not contained in any cluster is considered **noise**.



# DBSCAN Algorithm: Intuition

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“By recursively

exploring the neighborhood of **core points** within the  $\epsilon$ -distance

and incorporating **reachable points** into clusters,

the DBSCAN algorithm identifies dense regions in the dataset,

while also detecting **outliers** that do not fit within these dense regions.”

# DBSCAN Algorithm

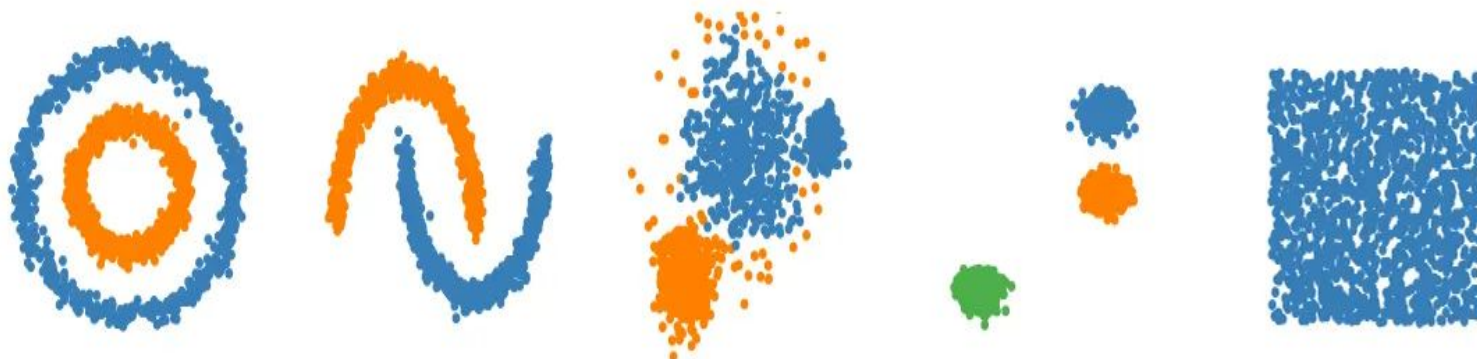
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1. Choose  $\epsilon$  (a positive number) and **MinPoints** (a natural number).
2. Select an arbitrary unvisited point **P** from the dataset.
3. Check if point **P** is a core point. If yes, form a cluster including **P**.
4. Recursively add density-reachable points within the  $\epsilon$ -neighborhood of the cluster points.
5. After fully expanding a cluster, select a new unvisited point and repeat the process **2** to **4**.
6. **Borders:** Assign border points to the cluster of a core point within their  $\epsilon$ -neighborhood.
7. **Noise Identification:** Label points that are neither core points nor border points as noise.

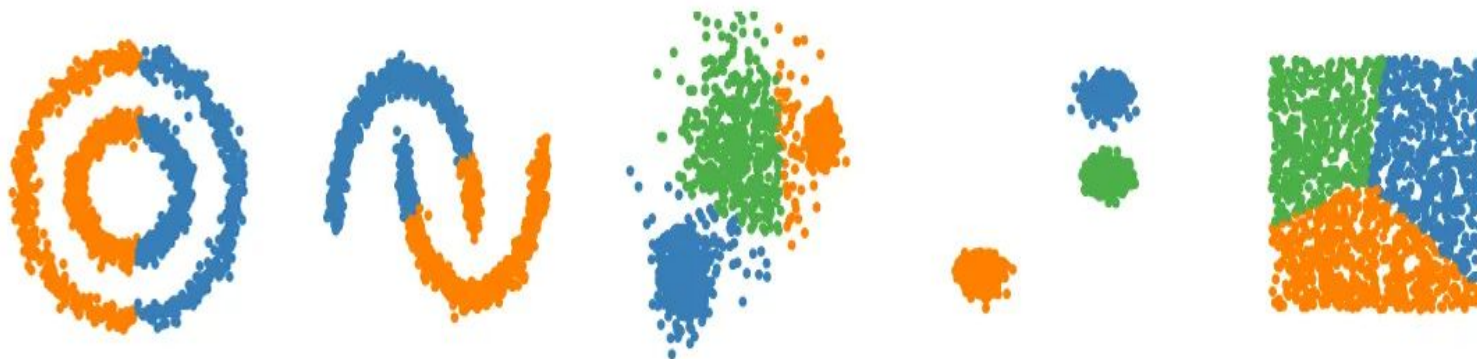
# DBSCAN vs K-Means (1 / 2)

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DBSCAN



k-means



# DBSCAN vs K-Means (2 / 2)

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- K-Means algorithm with different values of **K** and shapes of data:

<https://www.naftaliharris.com/blog/visualizing-k-means-clustering/>

- DBSCAN algorithm with different shapes of data:

<https://www.naftaliharris.com/blog/visualizing-dbscan-clustering/>

# How to Determine MinPts?

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General guidelines for setting **MinPts**:

- Larger datasets require a larger **MinPts** value.
- **MinPts** must be chosen at least 3.
- In noisier datasets, choose a larger **MinPts** value.
- **MinPts** should generally be  $\geq$  the dimensionality of the dataset.
  - **Example:**  $\text{MinPts} = 2 * \text{number of dimensions}$
- Domain knowledge is crucial to select an appropriate **MinPts** value.

# Determine $\epsilon$ using K-Distance Plot

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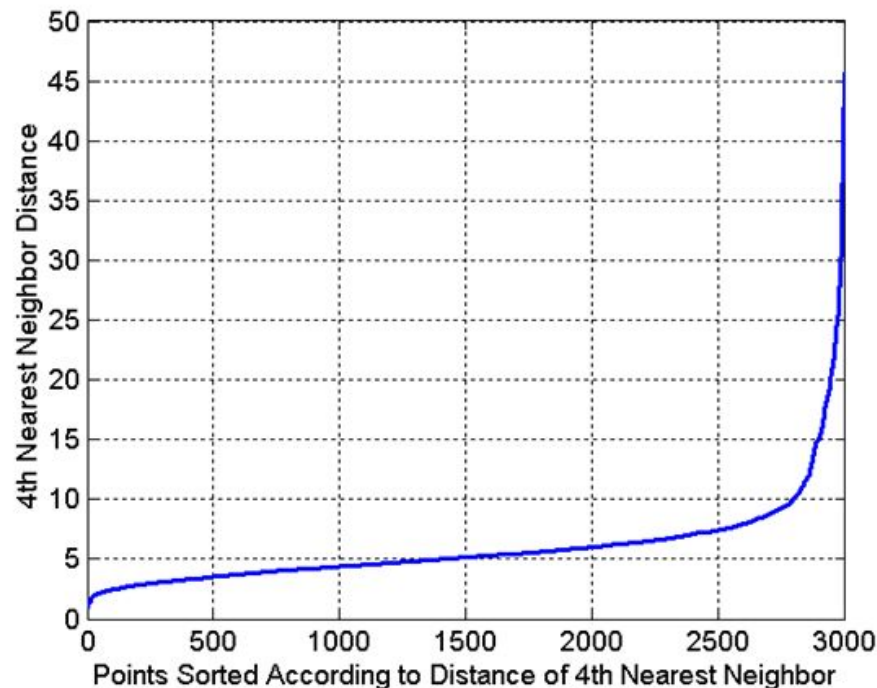
General guidelines for setting  $\epsilon$  using K-Distance Plot:

- Compute the distance of each point to its  **$K=MinPts-1$**  neighbors
- Plot these  **$K$** -distances in ascending order.
- The 'knee' in the plot indicates where a sharp change in distance occurs.
- This point is indicative of the optimal  $\epsilon$  value.
- Helps to distinguish between core, border, and noise points in the data.

# Determine $\epsilon$ using K-Distance Plot

- **Idea**: for points in a cluster, their **K**th nearest neighbors are at close distance
- Noise points have the **K**th nearest neighbor at farther distance.
- Plot sorted distance of every point to its **K**th nearest neighbor.

example  
with **MinPts=5**



# DBSCAN: Advantages and Disadvantages

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- **Advantages:**

- Can handle clusters of different shapes and sizes
- Resistant to noise and outliers
- Doesn't require predefined number of clusters.

- **Disadvantages:**

- Sensitivity to the two parameters  $\epsilon$  and **MinPts**
- Difficulty with varying density clusters
- Not suitable for high-dimensional data due to the curse of dimensionality